



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

Re - accredited by NAAC with 'A' Grade

Founder: Prof. Dr. S. B. Mujumdar, M.Sc., Ph.D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name : Supervised Machine Learning
Course Code : TE7499
Faculty : Engineering
Course Credit : 4
Course Level : 4
Sub-Committee (Specialization) : Computer Science

Learning Objectives :

1. Understand the concept of training set, testing, and validation set
- Perform the Exploratory Data Analysis
3. Apply the suitable algorithm based on problem
4. Apply the Regression algorithm for Prediction

Books Recommended :

Book	Author	Publisher
1. Machine Learning	Tom Mitchell	McGraw hill education 1st edition
2. The Machine Learning Engineering book	Andriy Burkov	Oliver

Course Outline :

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Introduction to Machine Learning, Supervised Learning, Unsupervised Learning, reinforcement learning, Applications, Training, Test, Validation data set	8	8
2	Regression Basics, Dependent variables, Independent variables, Correlation, Covariance, Ordinary Least Square method. Linear Regression with one Variable using Gradient Descent, Multiple Linear Regression using Gradient Descent & Normal Equation Method, Polynomial Regression, Evaluation Methodology and Evaluation Measures for Regression, Underfitting and Overfitting, Bias and Variance Trade off, Regularization Methods Ridge Regression, Lasso regression	20	20
3	Classification Linear Classifiers Logistic Regression, Binomial classification, Multiclass Classification Support vector machine Non-linear classifiers KNN, Bayes Rule and Naive Bayes Model, Decision Tree Classifier, random forests, Ensemble methods: Bagging, boosting : ADA boost, Gradient boost, XGboost Evaluation Metrics for Classification	20	20
4	Use Case for Regression	6	6
5	Use Case for Classification	6	6
Total		60	60

Pre Requisites :

None

Evaluation :

- A) Continuous Assessment (75 marks)
1. Essential
 - a) Quizzes b) Assignments c) Tests d) Viva-voce

B) End Semester Examination: N.A

Pedagogy :

1. Interactive Teaching Learning
2. Project based Learning
3. Hands-On

Expert :

Dr. Ketan Kotecha, Professor, SIT

Dr. Swati Ahirrao, Associate Professor (CS and IT), SIT